

We Forge the Chains We Wear in Life

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1 DESCRIPTION OF ARTWORK

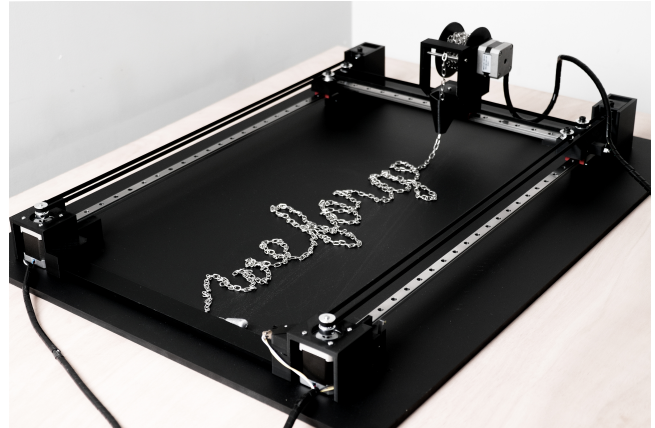
“We forge the chains we wear in life” is an aphorism by Charles Dickens. This chain is a metaphor for heavy and unbreakable things that bind our ideas and emotions. That is, words that tend to possess our minds and souls can also become chains. This inspired me to create a device that forms letters with chains.

Over the past decade, I have explored the possibility of displaying text and pictures using computer-controlled linear objects. Pixels, or dots, enable a wide variety of computer-generated images to be displayed. By contrast, humans use lines to write letters and draw pictures. Therefore, I believe that linear objects, such as strings and springs, can also be a medium for conveying people’s ideas and emotions, and the results of such exploration have been presented in previous SIGGRAPH conferences [Katsumoto 2018, 2023]. In this study, the focus is on the chains.

The device for lettering with chains is named Renment. This is because, in Japanese, renmen means to “continue endlessly without breaking,” and Japanese script typefaces are also called Renmen. The basic design of Renment involves a computer numerical control (CNC) milling machine and pen plotter. Two stepping motors installed on the frame move the XY position of the header, and another stepping motor mounted on the header feeds the chain from the reel. The characters’ thickness, tangling of lines, and blurring can be changed by controlling the speed at which the header moves and the amount of chains fed. Because letters are formed by chains, their shapes always fluctuate.

The control circuit uses Grbl [Jeon 2024], which were developed by volunteers and distributed under the GPLv3 license and Arduino CNC Shield V3.51. The mechanical design was initially based on an example from DIY Machines Ltd [DIY Machines Ltd. 2024]. Subsequently, all parts were recreated for this piece by adding a chain feed mechanism, timing belt tensioner, etc. The font was designed based on the alphabet of the Korean font Pilgiche after studying several existing legible script fonts, and it was G-coded for Grbl.

In this work, Renment automatically writes the abovementioned aphorism as the work title. When the device finishes writing, the



words are deconstructed, the letters return to the chains, and blanks remain, except scratches from a lead weight attached to the end of the chain remaining on the base. Then, the automatic writing begins again.

2 PROJECT URL

<https://www.katsumotoy.com/>
<https://sites.google.com/view/utsuroi/>

ACKNOWLEDGMENTS

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REFERENCES

- DIY Machines Ltd. 2024. Kinetic Sand Art Coffee Table - Self drawing. <https://www.diy-machines.co.uk/kinetic-sand-art-coffee-table-self-drawing>
- Sungeun(Sonny) K. Jeon. 2024. Grbl. <https://github.com/grbl/grbl>
- Yuichiro Katsumoto. 2018. Robotype: studies of kinetic typography by robot display for expressing letters, time and movement. In *ACM SIGGRAPH 2018 Art Gallery* (Vancouver, British Columbia, Canada) (SIGGRAPH '18). Association for Computing Machinery, New York, NY, USA, 386–393. <https://doi.org/10.1145/3202918.3203079>
- Yuichiro Katsumoto. 2023. Hinemosu 30. In *ACM SIGGRAPH 2023 Art Gallery* (Los Angeles, California) (SIGGRAPH '23). Association for Computing Machinery, New York, NY, USA, Article 5, 2 pages. <https://doi.org/10.1145/3588428.3593819>

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